

SOP-13: Enzymes and Antibiotics - Respiratory and Skin Sensitizers

Roberts Lab, School of Aquatic and Fishery Sciences - Rev. 0 - September 1, 2026

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): NO - Sections 9, 10 and 11 are optional; they are completed below as good practice.

Section 1 - Lab-Specific Information

Building/Room(s) covered by this SOP	Fisheries Teaching & Research (FTR) Building, Rooms 209 and 213
Unit or department	School of Aquatic and Fishery Sciences (SAFS), College of the Environment
Principal Investigator	Steven Roberts (work phone 206-685-3742)
PI signature / date	_____ Date: _____
This SOP was created by (if not PI)	_____ Date: _____
Emergency contact	9-1-1 (campus phone) EH&S 206-543-7262 (M-F 8-5) 206-685-UWPD (8973) after hours
Date of last review	_____ Next review due: _____ (review annually and after any incident or process change)

Chemicals covered by this SOP

Chemical name (as listed in MyChem)	CAS number
Liberase TM Research Grade (collagenase/thermolysin blend), 10 mg	not assigned / enzyme blend
HyClone Penicillin-Streptomycin Solution (SV30010)	mixture
HyClone Antibiotic/Antimycotic Solution (pen/strep/amphotericin B) (SV30079, SV3007901)	mixture
Deoxyribonuclease I (DNase I), 18047-019	9003-98-9
Lysyl endopeptidase (125-02543)	72561-05-8

Safety Data Sheets for every chemical listed above must be readily available in the laboratory. SDSs are accessible through MyChem.

Section 2 - Hazards

Applicable GHS pictograms: Health hazard (respiratory sensitization Cat 1); Exclamation mark (skin sensitization Cat 1, irritant)

- Stock chemicals: Liberase TM lyophilized enzyme blend (10 mg), penicillin-streptomycin solution (100 mL), antibiotic/antimycotic solution (200 mL and 1,000 mL), DNase I (100 microliters), lysyl endopeptidase (2 micrograms lyophilized). Intermediates and working solutions: reconstituted enzyme stocks, 1X antibiotic supplement in cell culture medium, DNase I working solutions in extraction and library prep protocols, and lysyl endopeptidase digests in

proteomics sample preparation. Wastes: unused enzyme and antibiotic solutions, spent culture medium, digest reactions, and contaminated labware.

- The defining hazard of this group is SENSITIZATION, not acute toxicity. Proteolytic enzymes (Liberase, lysyl endopeptidase, DNase I) are classified Respiratory sensitization Category 1 and Skin sensitization Category 1: repeated low-level exposure, especially to lyophilized powder dust, can induce occupational asthma or allergic dermatitis that then recurs at very low exposures. Beta-lactam antibiotics (penicillin) are among the most common causes of occupational and clinical allergy, and amphotericin B is a further sensitizer.
- Sensitization is not dose-dependent in the ordinary sense. Once a person is sensitized, an exposure far below any occupational exposure limit can trigger a reaction, and reactions can be severe. Prevention of the first sensitizing exposure is the entire control strategy.
- Routes of exposure: inhalation of lyophilized powder or aerosol (primary for sensitization), skin contact, eye contact, ingestion, and accidental injection.
- How exposure might occur: opening a vial of lyophilized enzyme and releasing a puff of powder; reconstituting enzyme with vigorous pipetting; aerosol from vortexing or pipetting; splash of antibiotic solution during medium preparation; and contact with contaminated gloves.
- Target organs: respiratory tract (occupational asthma, rhinitis) and skin (allergic contact dermatitis and urticaria). Penicillin allergy can additionally cause systemic reactions including, rarely, anaphylaxis.
- Signs and symptoms of exposure: sneezing, runny nose, itchy eyes, cough, wheeze or chest tightness that improves away from the laboratory and recurs on return - this pattern is the classic presentation of occupational sensitization and must be reported. Skin: redness, itching, hives or eczema on the hands and forearms. A person with a known penicillin allergy may react severely to the pen-strep and antibiotic/antimycotic solutions.
- GHS classification (from MyChem / SDS): Liberase TM - Respiratory sensitization Category 1; Skin sensitization implied; Serious eye damage/eye irritation Category 2A; Skin corrosion/irritation Category 2; STOT-single Category 3 (respiratory irritation). HyClone Penicillin-Streptomycin - Respiratory sensitization Category 1; Skin sensitization Category 1. HyClone Antibiotic/Antimycotic - the MyChem record shows NOT CLASSIFIED, but the product contains penicillin, streptomycin and amphotericin B and is handled here as a sensitizer regardless of that record. DNase I and lysyl endopeptidase - MyChem shows no identified hazard or not classified; both are proteolytic or nucleolytic enzymes and are handled here as respiratory sensitizers, which is standard practice for enzyme preparations.
- Note on inventory records: several products in this group are recorded in MyChem as 'NOT CLASSIFIED' or with no GHS categories. That reflects a gap in the inventory record, not an absence of hazard. This SOP applies the more protective classification. The PI should ask EH&S to update the MyChem records for these items.

Section 3 - Engineering Controls and Personal Protective Equipment (PPE)

Engineering controls

- Opening, weighing and reconstituting LYOPHILIZED enzyme (Liberase TM, lysyl endopeptidase) is performed inside a biological safety cabinet or the certified chemical fume hood in FTR 209, never on the open bench, because the dry powder is the principal sensitizing exposure. Any chemical fume hood used must be tested and passed by EH&S; the biological safety cabinet must be within its annual certification.
- Before opening a lyophilized vial, centrifuge it briefly or tap it down so that powder is not suspended in the headspace, and open it slowly inside the containment device.
- Reconstitute by adding liquid gently down the vial wall; do not pipette vigorously, vortex uncapped, or blow out the last drop, all of which generate aerosol.
- Antibiotic solutions and reconstituted enzyme stocks in liquid form may be handled at the bench or, preferably, in the biological safety cabinet when preparing culture medium.
- Closed-tube handling of working solutions is permitted at the bench in both FTR 209 and FTR 213.

- The nearest eyewash and safety shower are located in FTR 209 and FTR 213 and are inspected and flushed weekly by lab staff.

Hygiene measures

- Avoid contact with skin, eyes, and clothing. Wash hands after removing PPE, before breaks, and immediately after handling the chemical. If any chemical covered by this SOP comes into contact with any PPE, the PPE shall be immediately removed and discarded properly. Any potentially exposed body parts should be washed immediately.
- No eating, drinking, chewing gum, applying cosmetics, or storing food or drink in any area where these chemicals are used or stored.

Skin and body protection

- Chemically compatible laboratory coats that fully extend to the wrist must be worn, appropriately sized and buttoned to their full length. Personnel must also wear full-length pants, or equivalent, and close-toe shoes. The area of skin between the shoe and ankle must not be exposed.
- Lab coat type: buttoned barrier lab coat with cuffs covered by the glove. Because these materials are skin sensitizers, the goal is zero skin contact rather than tolerance of incidental contact.
- A disposable sleeve set is worn when reconstituting lyophilized enzyme.
- Any lab coat contaminated with enzyme or antibiotic solution is removed and laundered through the SAFS service; contaminated coats are not taken home, where they could sensitize household members.

Hand protection

- Hand protection IS required for the activities described in this SOP.
- Required gloves: 4-6 mil nitrile gloves for all handling of enzymes and antibiotic solutions, changed immediately on any contact and at the end of each task. Latex gloves are not used in this laboratory.
- Consult your preferred glove manufacturer to ensure that the gloves you plan to use are compatible with the specific chemical being used. Gloves must be inspected prior to use, including a check for pinholes.
- Remove gloves before touching door handles, keyboards, phones or your face; transferring sensitizer from glove to skin is a common route of sensitizing exposure.
- Use proper glove removal technique (without touching the glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands immediately after glove removal.

Eye protection

- ANSI Z87.1-compliant eye protection IS required for all work with these chemicals. Ordinary prescription glasses will NOT provide adequate protection unless they also meet the Z87.1 standard and have compliant side shields.
- Minimum eye protection: ANSI Z87.1 safety glasses with side shields for routine handling of liquid enzyme and antibiotic solutions. Chemical splash goggles are required when opening and reconstituting lyophilized enzyme and whenever there is a splash risk.

Respiratory protection

- Respiratory protection is NOT required for the activities described in this SOP, provided all lyophilized powder handling is confined to a biological safety cabinet or certified chemical fume hood as written.
- Lyophilized enzyme vials are NEVER opened outside of a containment device. Liquid antibiotic and enzyme solutions may be used outside of the fume hood at the bench or in the biological safety cabinet.

- Respiratory protection may be needed if a dust or aerosol hazard is present and work is conducted outside of the containment device; because such work is prohibited here, no respirator is needed. If any procedure may pose such a hazard, it should be eliminated or strictly isolated.
- Respirators should be used as a last line of defense (i.e., after engineering and administrative controls have been exhausted), and when any Action Limit (AL) or Occupational Exposure Limit (OEL) has been exceeded or when there is a possibility that an AL/OEL will be exceeded. If a potential exposure hazard cannot be eliminated, contact the EH&S Respiratory Protection Program administrator to discuss respiratory protection or to enroll in the program so a respiratory protection analysis can be performed. Program enrollment includes medical evaluation, training and fit testing for an appropriate respirator.
- An individual who becomes sensitized should not rely on a respirator to continue the work; the correct response is reassignment away from the sensitizer, arranged with the PI and EH&S.

Section 4 - Special Handling and Storage Requirements

- Substitution and minimization: purchase enzymes in solution rather than lyophilized where the supplier offers it, and purchase pre-supplemented medium rather than adding antibiotic concentrate, to eliminate the highest-exposure steps.
- Quantity limits: reconstitute the entire vial of lyophilized enzyme at once and aliquot into single-use tubes, so that the vial is opened only once in its life. Do not repeatedly open and re-weigh lyophilized material.
- Any individual with a known penicillin, streptomycin or amphotericin allergy must NOT handle the penicillin-streptomycin or antibiotic/antimycotic solutions. Inform the PI so that the task can be reassigned. No disclosure of medical detail is required beyond the fact that reassignment is needed.
- Any individual who develops respiratory or skin symptoms that improve away from the laboratory must report this to the PI and EH&S promptly and stop working with these materials pending evaluation.
- All containers - vials, aliquots, supplemented medium bottles and waste containers - must be labeled with the full name of the enzyme or antibiotic, the concentration, the word 'SENSITIZER', the date and the preparer's initials. Labeling supplemented medium matters: an unlabeled bottle of antibiotic-containing medium is how an allergic colleague gets exposed.
- Unattended operations: enzyme digests and cell cultures incubating in closed, labeled vessels are permitted; open handling is not left unattended.
- Storage: lyophilized enzymes and antibiotic solutions are stored at the temperature specified by the manufacturer (typically 2-8 degC or -20 degC) in labeled, closed secondary containers in the refrigerator or freezer in FTR 209 or 213, posted with a 'SENSITIZER' label. Segregate from unrelated reagents and from food-grade materials.
- Transport: vials and aliquots travel in closed, leak-proof secondary containers.
- Clean the biological safety cabinet or fume hood upon completion of tasks with a detergent solution followed by 70% ethanol; wipe dry, and record the cabinet decontamination in its log.
- Clean all contaminated surfaces with detergent solution followed by 70% ethanol and dry.
- Place all contaminated disposable items in appropriate laboratory waste for disposal, in a closed container.
- Non-disposable/re-usable utensils, glassware, and other surfaces contaminated with these materials must be decontaminated at the end of the laboratory work session.
- When work is completed, remove gloves and wash hands with soap and water.

Section 5 - Spill and Accident Procedures

Spill response

- Chemical spills must be cleaned up as soon as possible by properly protected and trained personnel. All other persons should leave the area. Do not attempt to clean up any spill if not trained or comfortable. Evacuate the area and call 9-1-1 on a campus phone for help.
- POWDER spill of lyophilized enzyme: do NOT brush, wipe dry, or use compressed air, all of which aerosolize a respiratory sensitizer. If the spill is contained inside the operating biological safety cabinet or fume hood, dampen it with water or 70% ethanol using a wash bottle, wipe up with damp absorbent working from the outside inward, and place the material in a closed waste container. Then wipe the surface with detergent solution and 70% ethanol.
- A lyophilized enzyme spill OUTSIDE a containment device: leave the area, close the door, allow at least 15 minutes for aerosol to settle, and then clean up wearing goggles, nitrile gloves and a lab coat using the damp method above, or contact EH&S if the quantity is more than a few milligrams or if anyone in the group is sensitized.
- LIQUID spill of antibiotic or enzyme solution - less than 100 mL: may be handled by trained lab staff wearing safety glasses, nitrile gloves and a lab coat. Absorb with absorbent pads working from the outside inward, place in a closed waste container, and wipe the area with detergent solution followed by 70% ethanol.
- Any liquid spill greater than 100 mL, or any spill in which a sensitized individual may have been exposed: restrict access, post a sign, and contact EH&S at 206-543-0467 during business hours or via 9-1-1 after hours.
- PPE required for cleanup: chemical splash goggles, nitrile gloves, buttoned lab coat with disposable sleeves, full-length pants and close-toe shoes.
- Signage and entry restrictions: post 'DO NOT ENTER - SENSITIZER SPILL' for a powder spill outside a containment device; do not allow a sensitized individual to enter until the area has been decontaminated.
- Spill area must be cleaned up in the following manner: after absorbent removal, clean the area thoroughly with detergent solution followed by clean water, then 70% ethanol, and dry.
- Spill cleanup materials must be disposed of in the following manner: double bag all absorbent and contaminated disposables, label with a UW Hazardous Waste Label naming the enzyme or antibiotic and 'spill debris', and submit a collection request to EH&S. Small quantities of dilute antibiotic-containing medium may be handled as described in Section 6.
- For questions on spill cleanup, contact EH&S spill consultants at 206-543-0467 during normal business hours (Monday-Friday, 8 a.m. to 5 p.m.).
- Any spill, exposure or near miss incident requires the involved person or supervisor to complete and submit an incident report on the EH&S website within 24 hours (certain types of incidents require immediate notification).

Exposure procedures - first aid

- Perform first aid immediately. Refer to the SDS for additional chemical-specific guidance.
- Skin exposure: wash the affected area with soap and running water for 15 minutes. Watch for redness, itching or hives. Seek medical evaluation for any skin reaction, and report it to the PI and EH&S.
- Eye exposure: use the eyewash for 15 minutes while holding the eyelids open; seek medical evaluation.
- Inhalation exposure: move the person out of the contaminated area to fresh air. If the person develops wheeze, chest tightness or difficulty breathing, call 9-1-1 - a sensitized individual can have a severe reaction. Get medical help for any respiratory symptom.
- Ingestion: rinse mouth with water; do not induce vomiting; get medical help.
- Sharps injury or accidental injection: scrub the exposed area thoroughly for 15 minutes using warm water and sudsing soap and seek immediate medical evaluation. Injection of penicillin-containing solution into an allergic individual is a medical emergency - call 9-1-1.
- A person known to be allergic to penicillin who has any contact with the pen-strep or antibiotic/antimycotic solutions should seek medical evaluation immediately and should not continue working.

- Get help. Call 9-1-1 or go to the nearest Emergency Department; provide details of exposure: agent (name the enzyme or antibiotic and state that it is a sensitizer), dose, route of exposure, and time since exposure. Bring the SDS and this SOP to the Emergency Department.
- Notify your supervisor as soon as possible for assistance.
- Notify EH&S immediately after providing first aid and/or getting help. During business hours (M-F, 8-5) call 206-543-7262. Outside of business hours call 206-685-UWPD (8973) to be routed to EH&S staff on call.
- Report any pattern of respiratory or skin symptoms that improve away from the laboratory to the PI and EH&S even if no single exposure event occurred; this is how occupational sensitization presents.

Section 6 - Waste Accumulation and Disposal Procedures

- Waste streams: (a) unused lyophilized enzyme and enzyme stock solutions; (b) unused antibiotic concentrate; (c) spent culture medium containing antibiotics; (d) digest reactions; (e) contaminated solid debris (vials, tips, tubes, plates, gloves).
- Antibiotic-containing waste should not be poured down the drain as a matter of stewardship: releasing antibiotics to the sewer contributes to environmental antimicrobial resistance. Collect antibiotic concentrate and bulk antibiotic-containing medium as chemical waste. For small volumes of spent culture medium that also contain biological material, follow the laboratory's biological waste procedure for the biological component and consult EH&S about the antibiotic component.
- Enzyme solutions are not acutely toxic but are sensitizers; collect them as chemical waste rather than sinking them, and keep contaminated debris in a closed container so that dried enzyme residue is not aerosolized later.
- Accumulate liquid waste at the point of generation in a sturdy, chemically compatible container with a securely closable screw-top lid, kept closed except when actively adding waste, in secondary containment.
- All chemical waste containers must be labeled with a UW Hazardous Waste Label listing every constituent and approximate percentage, starting from the moment the first drop of waste is added. Refer to 'How to Label Chemical Waste Containers' on the EH&S website.
- Solid debris is collected in a lined, CLOSED, labeled container - an open trash can holding dried enzyme residue is an inhalation exposure source.
- Manage chemical and hazardous chemical waste separately from other waste streams such as biohazardous waste. Never autoclave chemical waste because it can produce hazardous chemical vapors, aerosols, and explosive reactions.
- Cultures containing live cells or organisms plus antibiotics are a mixed waste; the biological component may require inactivation while the chemical component must not be autoclaved indiscriminately. Contact EH&S for routing before generating mixed waste.
- The University of Washington Environmental Health & Safety Department has full responsibility for collection of hazardous waste for the University, all its campuses, and off-site locations; University laboratories cannot contract with an outside vendor to collect hazardous waste.
- To request a collection of chemical waste, submit a form on the Chemical Waste Disposal webpage on the EH&S website or directly in MyChem inventory. Contact EH&S at 206-616-5835 or chmwaste@uw.edu with questions.
- Visit the Hazardous Material Disposal and Recycling webpage on the EH&S website for information on disposing, recycling and surplusing materials. Update the MyChem inventory when containers are emptied, received, transferred or surplused.

Section 7 - Protocol

- Procedure A - Reconstituting lyophilized enzyme (Liberase TM, lysyl endopeptidase). Bring the vial to room temperature inside a closed secondary container. Inside the biological safety cabinet or certified fume hood, wearing

goggles, nitrile gloves, lab coat and disposable sleeves: centrifuge or tap the vial to settle the powder, open it slowly, add the specified volume of sterile buffer or water gently down the vial wall, cap, and allow it to dissolve by standing or gentle inversion - do not vortex uncapped. Aliquot into single-use labeled tubes and freeze. Discard the empty vial and tips into a closed waste container. Wipe the cabinet with detergent solution and 70% ethanol.

- Procedure B - Preparing antibiotic-supplemented culture medium. In the biological safety cabinet, wearing nitrile gloves, safety glasses and a lab coat, add penicillin-streptomycin or antibiotic/antimycotic concentrate to medium at the specified dilution using a serological pipette; avoid splashing and do not blow out the last drop. Label the medium bottle with 'CONTAINS PENICILLIN/STREPTOMYCIN - SENSITIZER', the date and initials.
- Procedure C - Tissue dissociation with Liberase TM. Thaw one single-use aliquot, add it to the dissociation buffer in a capped tube inside the biological safety cabinet, and incubate. Keep tubes capped during agitation. Collect the spent dissociation buffer as chemical waste.
- Procedure D - DNase I treatment and lysyl endopeptidase digestion. Use single-use aliquots of reconstituted enzyme in capped tubes at the bench with nitrile gloves, safety glasses and a lab coat. Collect completed reactions as chemical waste.
- Prohibited: opening lyophilized enzyme vials outside a containment device; vortexing uncapped enzyme solutions; handling of penicillin-containing solutions by an allergic individual; leaving contaminated debris in an open trash can; use by untrained personnel.
- Refer to Section 2 of the UW Laboratory Safety Manual on the EH&S website for additional guidance on chemical management and preparation for use.
- NOTE: Any deviation from this SOP requires approval from the Principal Investigator.

Section 8 - Special Precautions for Animal Use

No. N/A - these materials are used in cell culture and in sample preparation from euthanized animals, and are not administered to live animals under this SOP. If antibiotics are ever administered to live animals, the use will be documented and approved by IACUC and this SOP revised beforehand.

IACUC protocol number (if applicable): _____

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): NO - Sections 9, 10 and 11 are optional; they are completed below as good practice.

Section 9 - Approvals Required

- All staff working with the enzymes and antibiotics covered by this SOP must be trained on this SOP prior to starting work. They must also review the SDS for each product, and the SDSs must be readily available in the laboratory (MyChem). All training must be documented and maintained by the PI or their designee.
- Prerequisite training, all documented before independent work: (1) EH&S Laboratory Safety Training, initial and current refresher; (2) EH&S Managing Laboratory Chemical Waste training; (3) documented review of this SOP and the SDSs; (4) documented instruction on sensitization - what it is, how it presents, and the requirement to report symptoms; (5) biological safety cabinet use training where the cabinet is used.
- A worker must complete at least one supervised run of Procedure A, observed and signed off by the PI or a designated experienced lab member, before reconstituting lyophilized enzyme independently.
- Before assignment to work with the penicillin-streptomycin or antibiotic/antimycotic solutions, each individual is asked to confirm that they do not have a known allergy to those agents. No medical detail is requested or recorded; an individual who indicates a concern is reassigned. This is an authorization check, not a medical examination.

- Any individual who develops signs of sensitization must stop this work and be evaluated before returning to it. Reassignment away from the sensitizer, not additional PPE, is the correct control.
- No medical examination is required for the work described because respirators are not used.
- The PI should request that EH&S update the MyChem hazard records for the products in this group that are currently listed as NOT CLASSIFIED, so that the inventory reflects their sensitizer status.

Section 10 - Decontamination

- Biological safety cabinet: wipe the work surface and interior walls with detergent solution, then 70% ethanol, at the end of each session; record the decontamination in the cabinet log. Do not use bleach on the cabinet's stainless surfaces without following the cabinet's rinse procedure.
- Fume hood: wipe the work surface with detergent solution then 70% ethanol; remove and dispose of any liner.
- Equipment: wipe pipettors, racks and incubator surfaces with detergent solution then 70% ethanol. Enzyme residue that dries on a surface can be re-aerosolized, so surfaces are wiped damp rather than allowed to dry with residue on them.
- Glassware and reusable bottles: rinse into the appropriate waste container, then wash with detergent and hot water and rinse three times with deionized water. Proteolytic enzyme residue is removed by detergent and heat.
- Controlled areas: there are no glove boxes or restricted-access hoods in the Roberts Lab space; the biological safety cabinet and the fume hood are the controlled areas for this SOP.
- Cleaning solutions and materials: laboratory detergent (Alconox or Contrad) solution, hot water, 70% ethanol, lint-free wipes.

Section 11 - Designated Area

- Reconstitution of lyophilized enzyme: the biological safety cabinet or the certified chemical fume hood in FTR 209, posted 'SENSITIZER - DO NOT OPEN LYOPHILIZED VIALS OUTSIDE THIS CABINET'.
- Preparation of antibiotic-supplemented medium and cell culture work: the biological safety cabinet in FTR 209.
- Closed-tube handling of working solutions: the molecular biology benches in FTR 209 and FTR 213.
- Waste containers for these materials are kept closed within the designated areas.
- Storage: labeled closed secondary containers in the refrigerator and freezer in FTR 209 and FTR 213, posted with a 'SENSITIZER' label.

Section 12 - Documentation of Training

- Prior to using substances included in this SOP, laboratory personnel must be trained on the hazards described in this SOP, how to protect themselves from the hazards, and emergency procedures.
- Ready access to this SOP and to a Safety Data Sheet for each hazardous material described in the SOP must be made available in the lab space(s) where these substances are used.
- The Principal Investigator, or Responsible Party if the activity does not involve a PI, must ensure that their laboratory personnel have attended appropriate laboratory safety training (and refresher training where applicable).
- Training must be repeated following any revision to the content of this SOP.
- Training must be documented. This training sheet is provided as one option; other forms of training documentation (including electronic) are acceptable but records must be accessible and immediately available upon request.

I have read and understand the content of this SOP:

[illegible]